

Efforts are en route to eradicate the requisite knowledge of English for using computers and IT all over India

when one is using a regular tablet for input. The keyboard as an input solution for each of these characters is inappropriate because to be able to press the right combination, one needs to be trained. And trainers are not easily available.

The Problem

The picture about future PC usage and IT penetration in India looks rosy from the numbers available: they're on the upswing. But the numbers are nowhere near what they could have been if not for the language barrier. Indeed, one of the key reasons for the relatively low IT penetration rate in India is linguistic diversity.

Today, a number of solutions are available to surmount the problem of computer use by non-English-speaking people. The most common is stickers imprinted with the alphabet of a particular Indian language to be overlaid on the keyboard. But like we said, even with stickers, training is necessary.

As an inelegant solution, some use Indian-language translation software. A number of translation software are available on the Web: search on the term "Hindi translation software", and you'll get over a million results.

One Step Forward

A recent innovation is a gesture-based keyboard designed by HP Labs India, Bangalore. This keyboard was the runner-up in the Consumer Electronics category in the 2006 Technology Innovation Awards of *The Wall Street Journal*. The research team behind the design of the keyboard was led by Shekhar Borgaonkar, Department Manager—Access Devices, HP Labs India, who worked with eight other engineers.

The Gesture-based Keyboard is an electronic tablet with pen-based input on a touch-sensitive pad, where letters of Indian languages are entered by gestures. The Gesture-based Keyboard technology has been patented by HP, and is test-marketed by Prodigy Labs India Pvt Ltd, which offers solutions in the embedded technology space. Prodigy Labs is marketing the keyboard under the name *uRekha*.

How It Works

uRekha currently supports Hindi, Kannada, and Malayalam. For support of other Indian languages, it requires just about a month's time, according to Borgaonkar. We can soon expect the Gesture-based Keyboard to support several languages.

The *uRekha* package has a graphics tablet, a special pen, and the installation disc. It is priced at Rs 2,500. The surface of the tablet has consonants, vowels, and numbers imprinted.

Using the pen to stroke and tap on the characters on the graphics tablet will display that letter on the screen. Say you want to write an

Indian name, "Ashok", in Hindi. The consonants in "Ashok" are first tapped on the tablet.

In some detail, using the pen, after the "v" in "Ashok", the first consonant—"k"—is tapped on the tablet. After adding "k", you need to modify it to "ks". For that, you draw the vowel modifier "KS" on the consonant. The "d" does not need to be modified, so the resultant word would contain both non-modified and modified consonants.

When the gesture to input a vowel modifier on a consonant is performed, the pen electronically captures the gesture and the trajectory of the gesture. The vowel modifier is rendered using this information. Of course, for only consonants or vowels, the pen is simply to be tapped on the character.

To watch a demo of how *uRekha* works, take a look at the Flash demo on this month's DVD under the *Digital Leisure* section. The device can be used with Windows XP with Office 2002 or higher, and on Linux as well.

Who's Using It

uRekha is currently being used in Karnataka, Kerala, and also in the other south Indian states. "We did a pilot study in certain areas and thereafter framed marketing strategies. We're doing road shows and demonstrations for promotion and to inform people about *uRekha*. So far, we've been successful in selling more than 500 units—200 were sold to the government of Karnataka, 100 to the government of Tamil Nadu, and the others to individual customers," informs Harsha K V, Managing Director, Prodigy Labs, an Embedded technology company involved in distribution of *uRekha*.

Government employees are currently using it as a data entry input tool for documents, files, and reports. *uRekha* is also being used for filling forms currently in Hindi, Kannada, and Malayalam).

There's More...

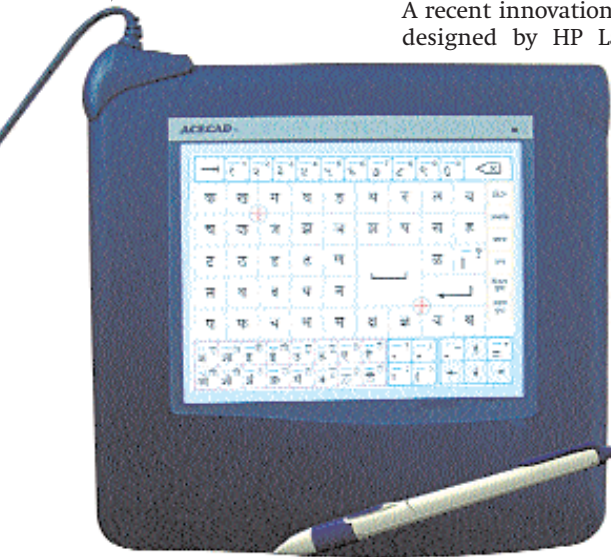
There is more and more research and development to break the language barrier in working with computers being done. A sizable number of organisations and academic institutions—CDAC (Centre for Development of Advanced Computing), to name one—are involved in the development of products and services in Indian languages. For instance, in the recent past, an Indian portal, Guruji.com, announced a search engine in Hindi, Kannada, and Telugu. Also, a number of OCR (Optical Character Recognition) software are available for non-English languages as well.

The Department of Information Technology has initiated the Technology Development for Indian Languages (TDIL), which comes out with interactive CDs and other information-processing tools in Indian languages.

"Currently, a real-life study is being done on the usage and popularity of the Gesture-based Keyboard. We also have plans to bundle it with our computers," says Borgaonkar.

Language could very well no more be a barrier for entry into the world of computing and information technology. Progress towards the availability of the necessary tools has been slow, but it's happening. And that's heartening. ■

samir_makwana@thinkdigit.com



uRekha will make typing in Hindi so much easier